

Academic Year	Module	Assessment Number	Assessment Type
2025	6CS007 – Project and Professionalism	2	Individual Report

Artifact Design – Pustakyatra

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Artifact

An artifact is a deliverable produced during the software development process. It can include diagrams, models, documents, wireframes, backlogs, and test plans that help developers understand how a system should be designed and implemented. Artifacts play an important role in guiding the development process by clearly defining system requirements and expected behavior. Without artifacts, software development becomes unclear and unstructured, similar to constructing a building without architectural plans.

For this project, Pustakyatra - a Digital E-Book Reader Platform, artifacts are used to ensure clarity in system design, development flow, and feature implementation. These artifacts help align development activities with user requirements and project objectives.

The project follows the Agile methodology for software development. Agile is selected because it supports iterative development, where features are developed, tested, and improved continuously. Unlike the Waterfall model, which follows a linear and rigid process, Agile allows flexibility and quick adaptation to changes based on user feedback.

Agile methodology encourages continuous collaboration between developers and stakeholders, helping to maintain software quality throughout the development lifecycle. Development and testing activities are carried out simultaneously, reducing the chances of major errors at later stages. This approach is suitable for Pustakyatra as new features such as user authentication, book catalog management, and reading functionality can be added incrementally.

Before implementation, wireframes are created as an initial artifact to visualize the user interface and improve user experience. Wireframes help identify design improvements early and reduce development rework.

Product Backlog

The product backlog is a prioritized list of features and requirements that need to be developed in the system. It represents all functionalities required for the Pustakyatra digital e-book reader platform and is continuously updated throughout the development process.

Actor	Product Backlog	Priority
Admin	Admin can log in and manage the platform	Must
Admin	Admin can manage users	Must
Admin	Admin can add, update, and delete books with images	Must
Admin	Admin can manage rentals and expiries	Must
Author	Author can upload and manage their books	Must
Author	Author can view feedback on uploaded books	Should
User	User can register, log in, and log out	Must
User	User can browse books and view book images	Must
User	User can rent books for a limited time	Must
User	User can purchase books permanently	Must
User	User can read books online	Must
User	User can bookmark, highlight, and add notes	Should
User	User can rate and review books	Should
System	System processes payments securely	Must
System	System sends rental expiry notifications	Should

Table 1: Product Backlog of Pustakyatra

Sprint Backlog – Pustakyatra

Sprint	Actor	Sprint Task
Sprint 1	User	User can register an account
Sprint 1	User	User can log in and log out securely
Sprint 1	Author	Author can register and log in
Sprint 1	Admin	Admin can log in to the admin dashboard
Sprint 1	Admin	Admin can manage user accounts
Sprint 2	User	User can browse books and view book cover images
Sprint 2	User	User can rent books for a limited time
Sprint 2	User	User can purchase books permanently
Sprint 2	Author	Author can upload books with details and files
Sprint 2	Author	Author can update or delete uploaded books
Sprint 2	Admin	Admin can add, update, and delete books with images
Sprint 2	System	System processes payments securely
Sprint 3	User	User can read books online
Sprint 3	User	User can save reading progress
Sprint 3	User	User can bookmark, highlight, and add notes
Sprint 3	User	User can rate and review books
Sprint 3	Author	Author can view reviews and feedback
Sprint 3	System	System sends rental expiry notifications

Table 2: Sprint Backlog of Pustakyatra

Functional Decomposition Diagram (FDD) – Pustakyatra

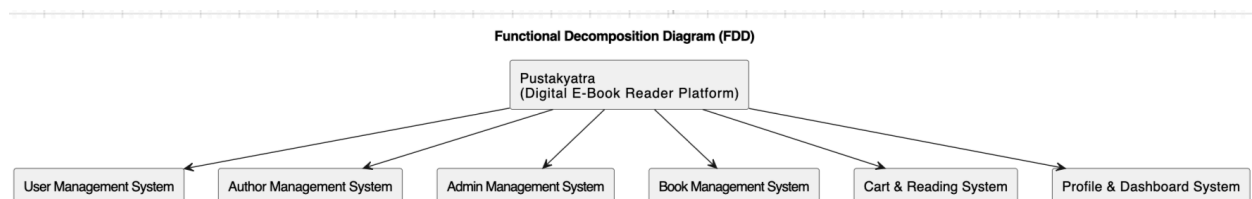


Figure 1: Functional Decomposition Diagram of Pustakyatra

The Functional Decomposition Diagram (FDD) illustrates the overall structure of the Pustakyatra system by breaking the main system into smaller functional subsystems. These subsystems represent the major functionalities required to operate the digital e-book platform effectively.

Legend (for Pustakyatra)

AMS – Admin Management System
UMS – User Management System
AuMS – Author Management System
BMS – Book Management System
CRS – Cart & Reading System

Requirement Types:

- FR – Functional Requirement
- NFR – Non-Functional Requirement
- UR – Usability Requirement

Admin Management System (AMS)

Req. Code	Requirement Description	Use Case	MoSCoW Prioritization
AMS-FR-1.0	The system allows the admin to login.	Admin Login	Must Have
AMS-FR-2.0	The system allows admin to create new admin accounts.	Create Admin	Must Have
AMS-FR-3.0	The system allows admin to delete other admins.	Delete Admin	Must Have
AMS-FR-4.0	The system allows admin to view all admins.	View Admins	Should Have
AMS-FR-5.0	The system allows admin to logout securely.	Admin Logout	Must Have
AMS-NFR-2.1	Admin actions should be secure and authenticated.	Security Control	Must Have
AMS-UR-1.0	Admin dashboard should be easy to navigate.	Dashboard UI	Should Have

Table 3: SRS of Admin Management System

Activity Diagram 1 – Admin Login

Activity Diagram - Admin Login (AMS)

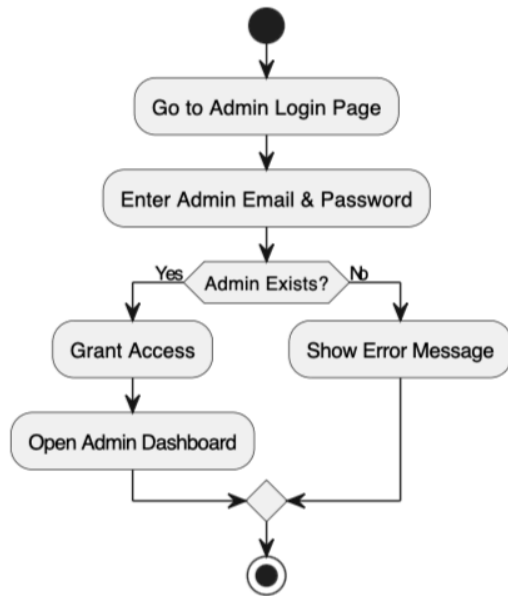


Figure 2: Activity Diagram 1 of Admin Management System

Activity Diagram 2 – Create / Delete Admin

Activity Diagram - Create/Delete Admin (AMS)

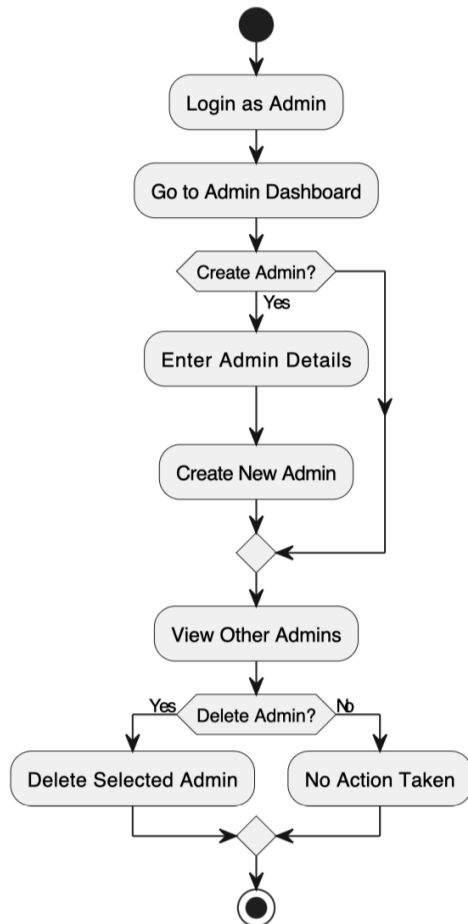


Figure 3: Activity Diagram 2 of Admin Management System

The activity diagrams illustrate the workflow of the Admin Management System in Pustakyatra. The first diagram represents the admin login process, ensuring secure authentication before accessing the dashboard. The second diagram demonstrates the admin's ability to create, view, and delete other admin accounts, highlighting system control and administrative authority.

Use Case Diagram – Admin Management System (AMS)

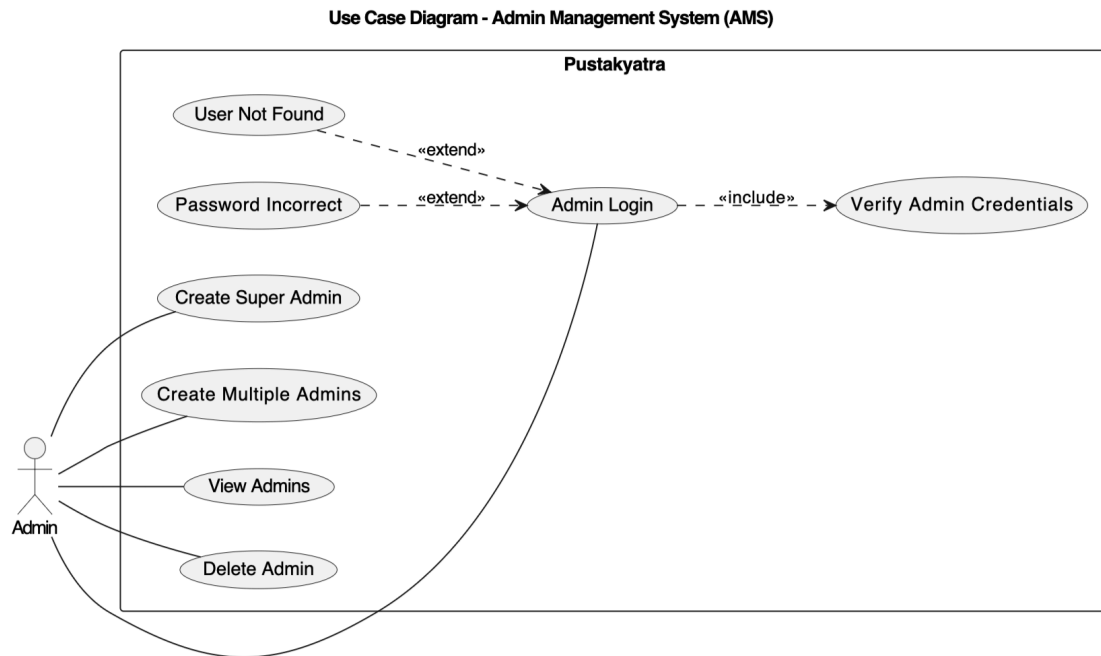


Figure 4: Use Case Diagram for Admin Management System (AMS)

Sequence Diagram – Admin Management System (AMS)

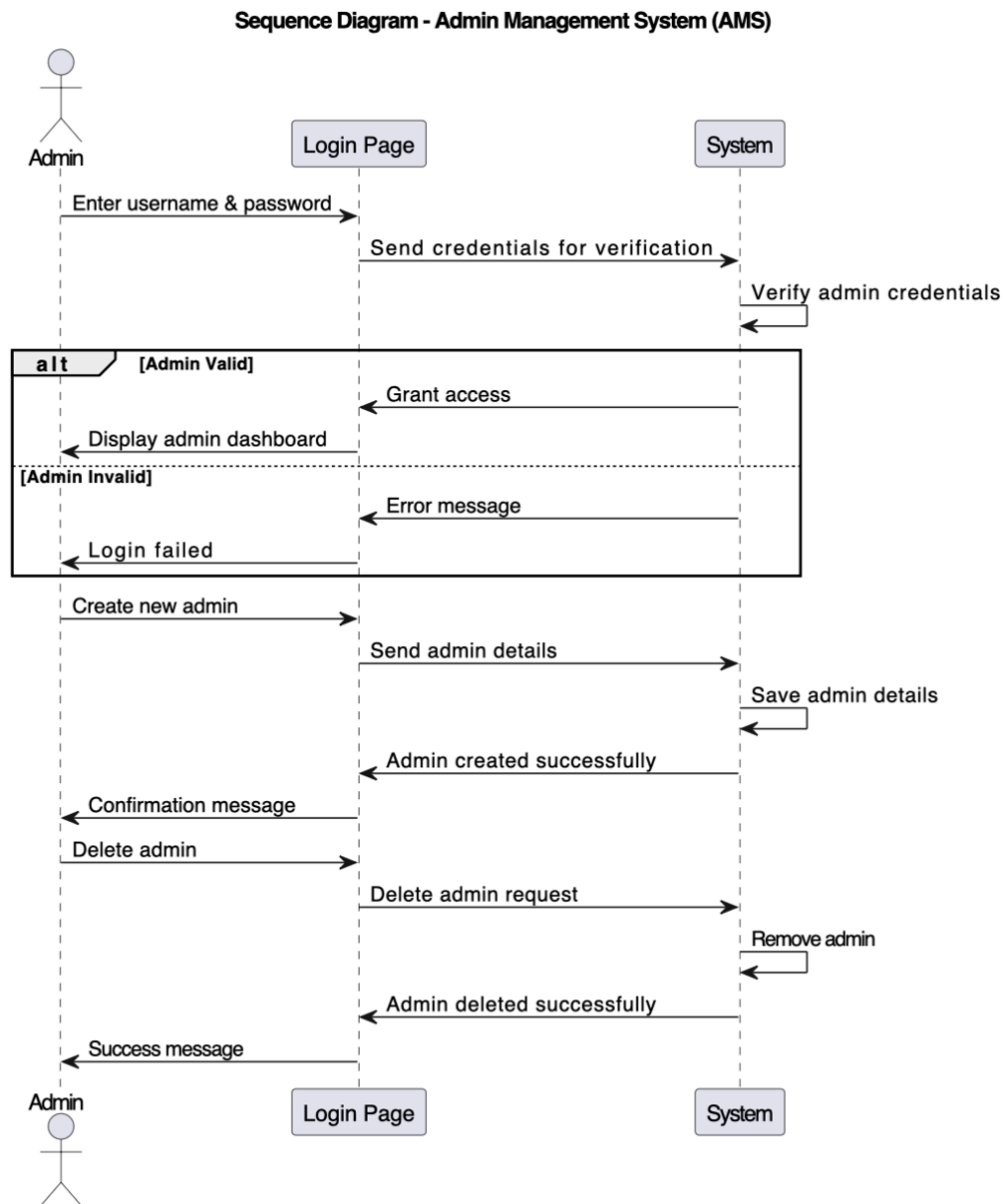


Figure 5: Sequence Diagram for Admin Management System (AMS)

The use case diagram illustrates the interactions between the admin and the Admin Management System in Pustakyatra. It highlights core administrative functionalities such as login, creation, viewing, and deletion of admin accounts, including exception scenarios such as

invalid credentials. The sequence diagram further explains the step-by-step interaction between the admin and the system, showing how requests are processed and responses are generated during login and admin management operations.

User Management System (UMS)

Req. Code	Requirement Description	Use Case	MoSCoW Prioritization
UMS-FR-1.0	The system shall allow the user to sign up and log in using email and password.	User Signup & Login	Must Have
UMS-FR-1.1	The system shall allow users to sign up and log in using social media (Google).	Social Login	Could Have
UMS-NFR-1.2	User credentials must be encrypted for security purposes.	Secure Authentication	Must Have
UMS-UR-1.3	The password must contain alphabets, numbers, and special characters.	Password Validation	Could Have
UMS-FR-2.0	The system shall allow users to reset their password if forgotten.	Reset Password	Should Have
UMS-UR-2.1	The system shall display an alert after successful password reset.	Password Reset Alert	Should Have
UMS-FR-3.0	Logged-in users shall be able to view all accessible pages.	View Pages	Must Have
UMS-FR-4.0	Admin shall be able to add and delete users.	Manage Users	Must Have

Table 4: SRS for User Management System (UMS)

Activity Diagram 1 – User Signup & Login (UMS)

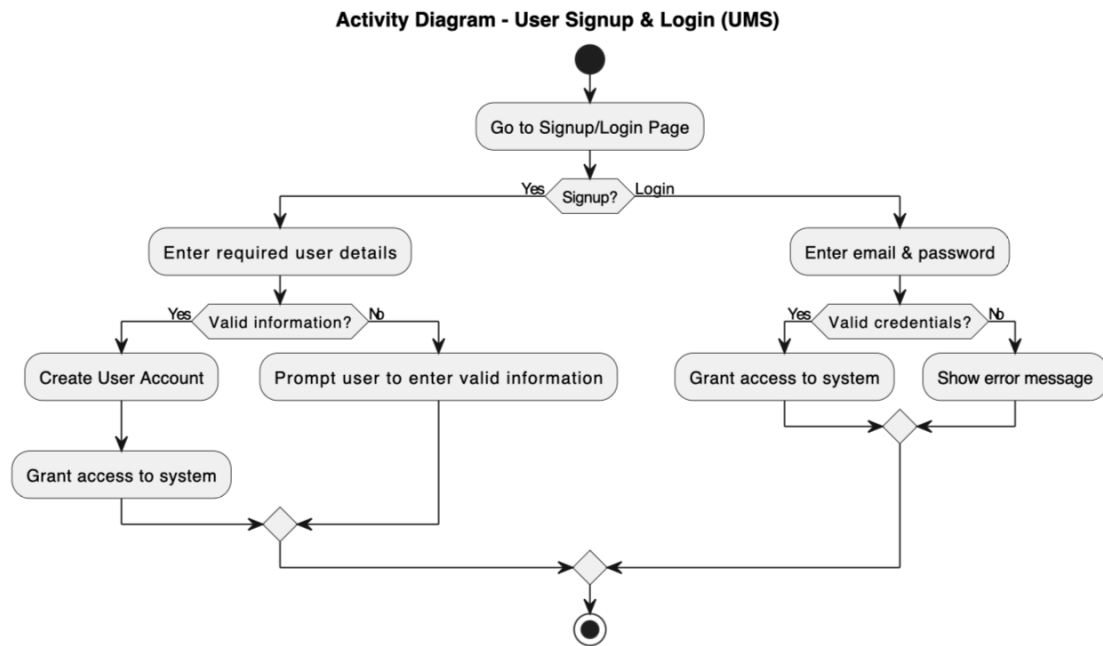


Figure 6: Activity Diagram 1 for UMS

Activity Diagram 2 – Password Reset (UMS)

Activity Diagram - Password Reset (UMS)

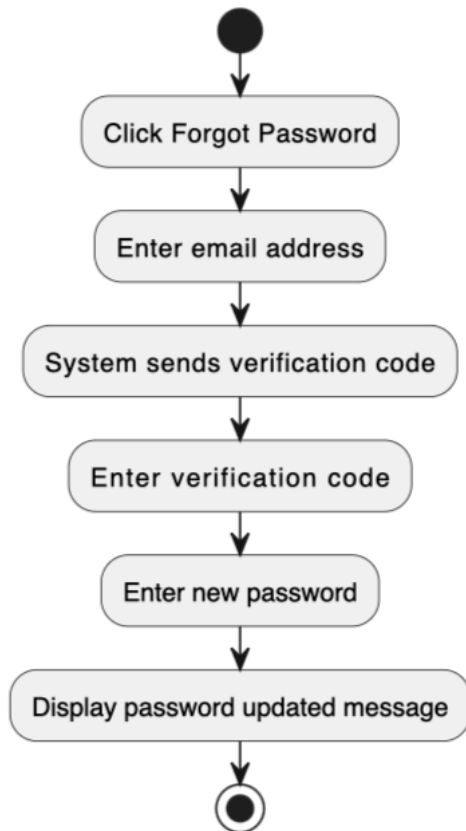


Figure 7: Activity Diagram 2 for UMS

Activity Diagram 3 – Admin Managing Users (UMS)

Activity Diagram - Admin Manage Users (UMS)

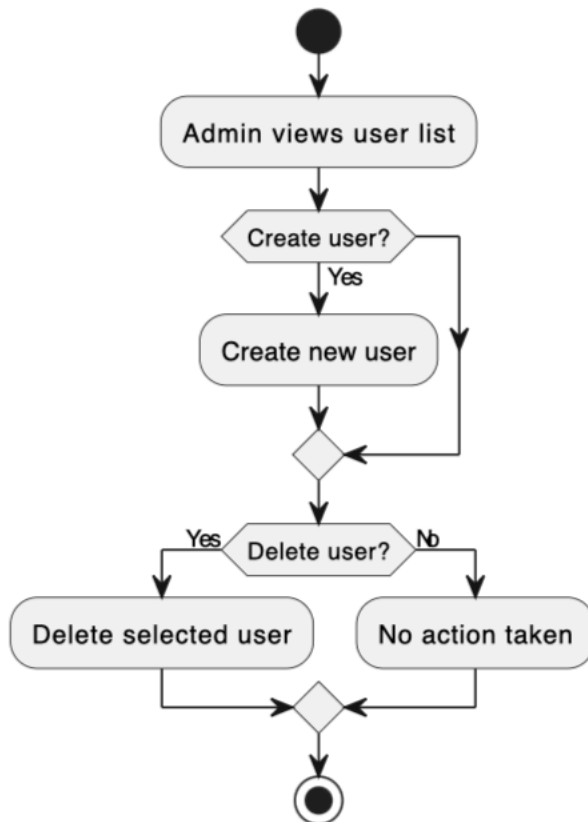


Figure 8: Activity Diagram 3 for UMS

Use Case Diagram – User Management System (UMS)

Use Case Diagram - User Management System (UMS)

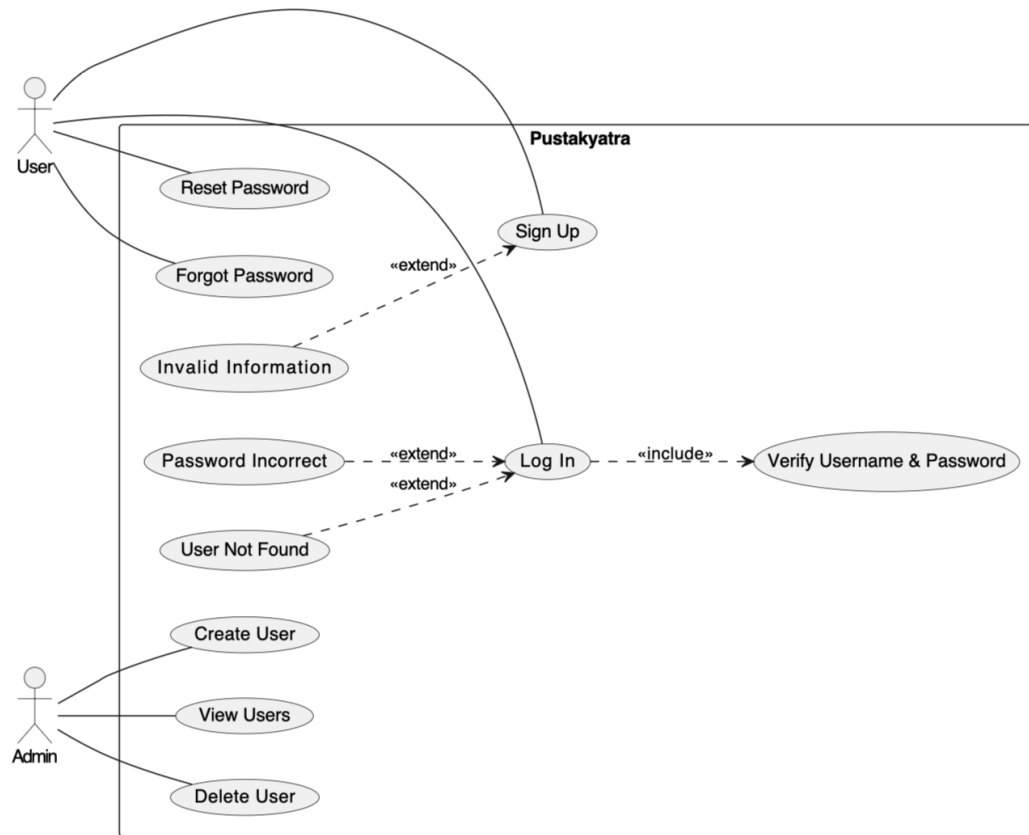


Figure 9: Use Case Diagram for UMS

Sequence Diagram – User Management System (UMS)

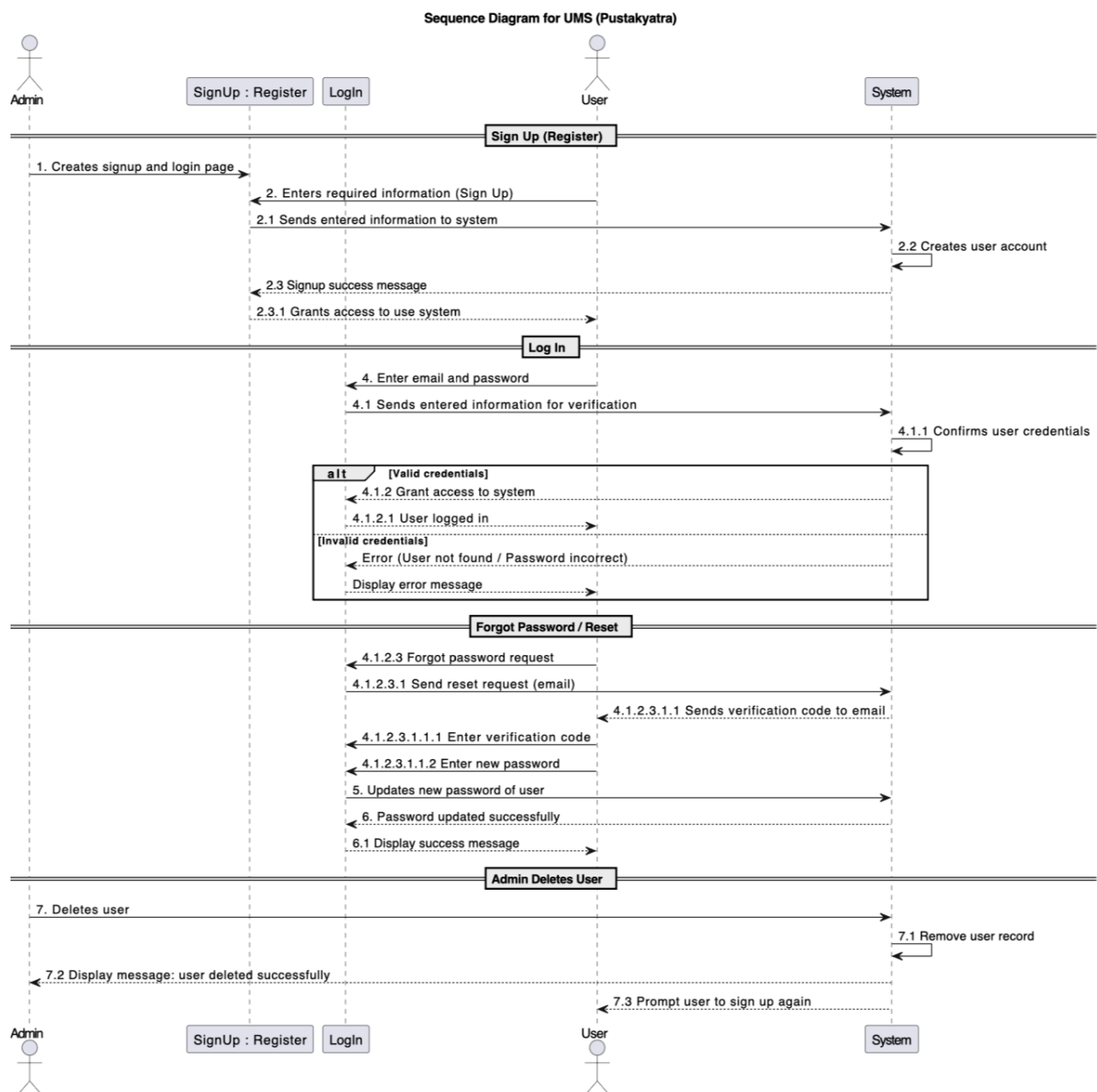


Figure 10: Sequence Diagram for UMS

The User Management System (UMS) handles user authentication, registration, password recovery, and user access control in Pustakyatra. Activity diagrams illustrate the workflow for

signup, login, password reset, and administrative user management. The use case and sequence diagrams further demonstrate user-system interactions, ensuring clarity, security, and usability.

CMS (Comment/Discussion Management System)

Req. Code	Requirement Description	Use Case	MoSCoW
CMS-FR-1.0	The system shall allow a reader to add a review for a book (text + date).	Add Review	Must Have
CMS-FR-1.1	The system shall allow a reader to edit or delete their own review.	Edit/Delete Review	Must Have
CMS-FR-1.2	The system shall display all reviews for a book to other users.	View Reviews	Must Have
CMS-FR-2.0	The system shall allow a reader to rate a book (1–5 stars).	Add Rating	Must Have
CMS-NFR-2.1	The system shall calculate and display average rating for each book.	View Avg Rating	Must Have
CMS-FR-3.0	The system shall allow readers to like/unlike a review (helpful).	React to Review	Could Have
CMS-NFR-3.1	The system shall show number of likes on each review.	View Review Reactions	Could Have
CMS-FR-4.0	Admin shall be able to delete inappropriate reviews.	Moderate Reviews	Should Have

Table 5: SRS for Comment/Discussion Management System (CMS)

2) Activity Diagram 1 — Add / Edit / Delete Review

Activity Diagram 1 - Review Management (CMS) - Pustakyatra

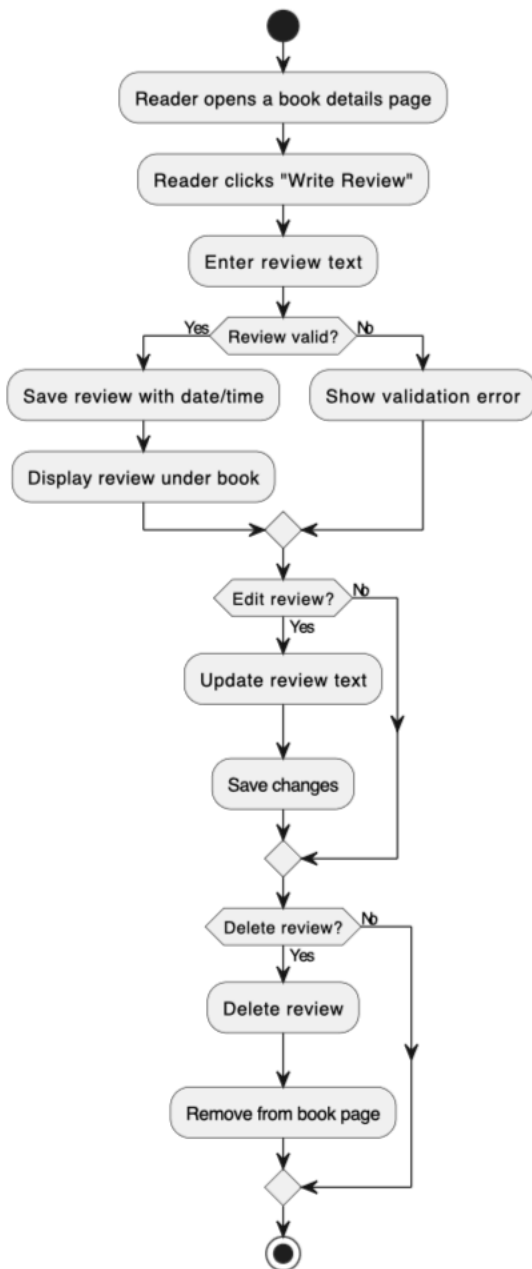


Figure 11: Activity Diagram 1 for CMS

Activity Diagram

Activity Diagram 2 - Rating Management (CMS) - Pustakyatra

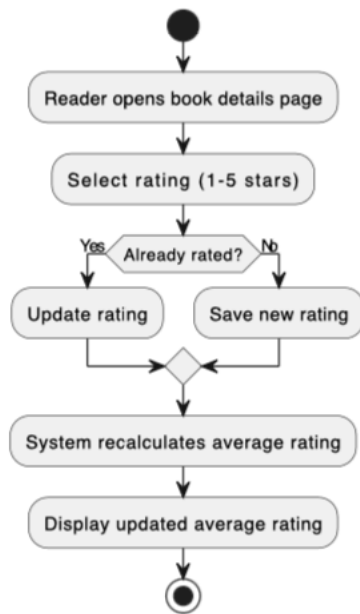


Figure 12: Activity Diagram 2 for CMS

Activity Diagram

Activity Diagram 3 - Admin Moderation (CMS) - Pustakyatra

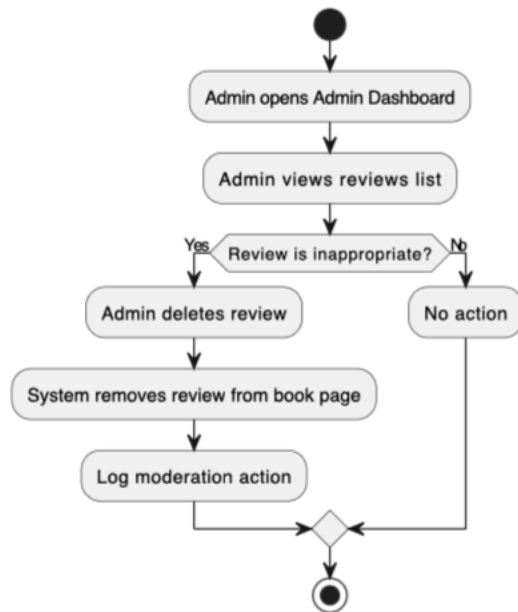


Figure 13: Activity Diagram 3 for CMS

Use Case Diagram – CMS

Use Case Diagram - CMS (Reviews & Ratings) - Pustakyatra

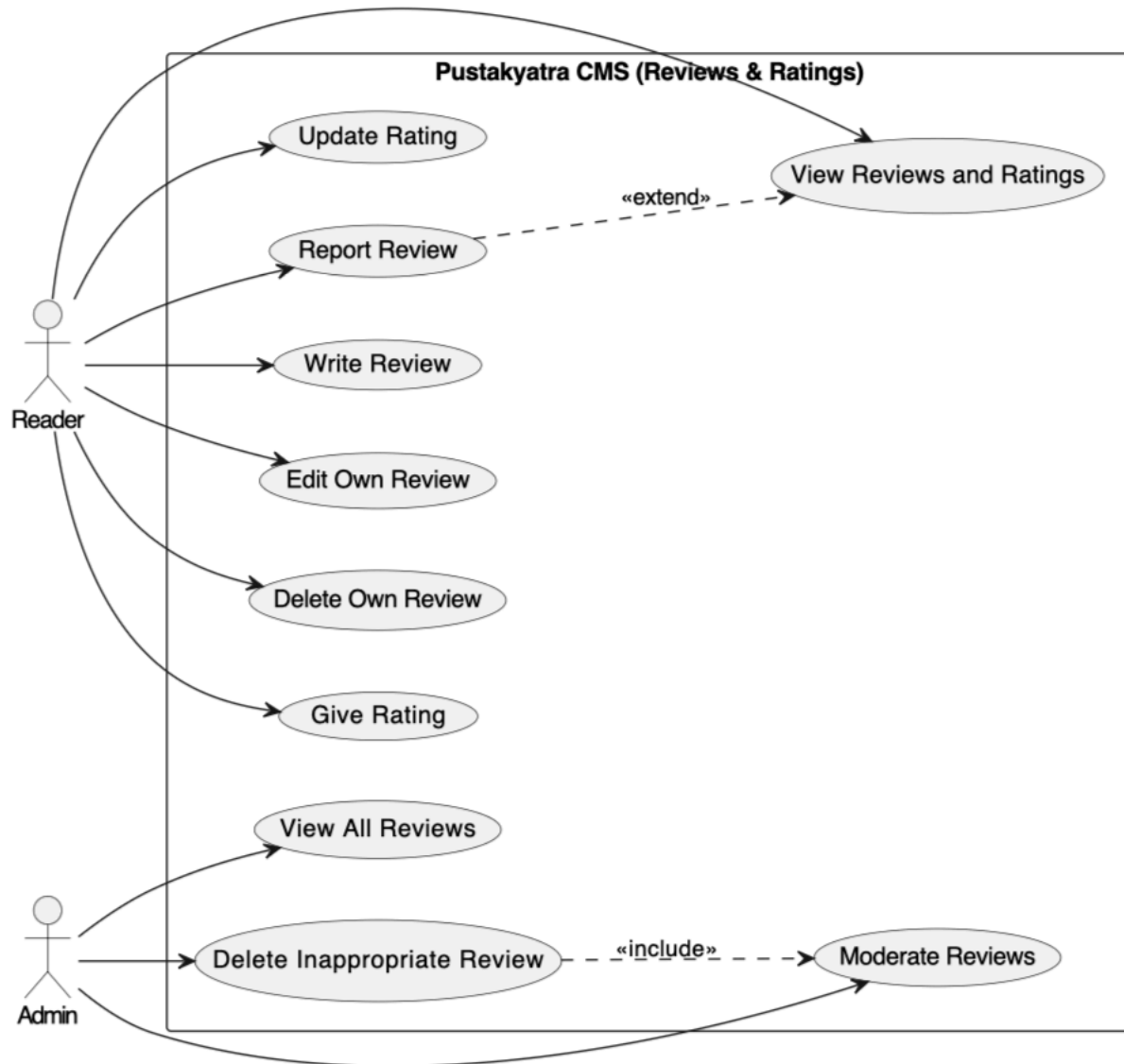


Figure 14: Use Case Diagram for CMS

Sequence Diagram – CMS (Pustakyatra)

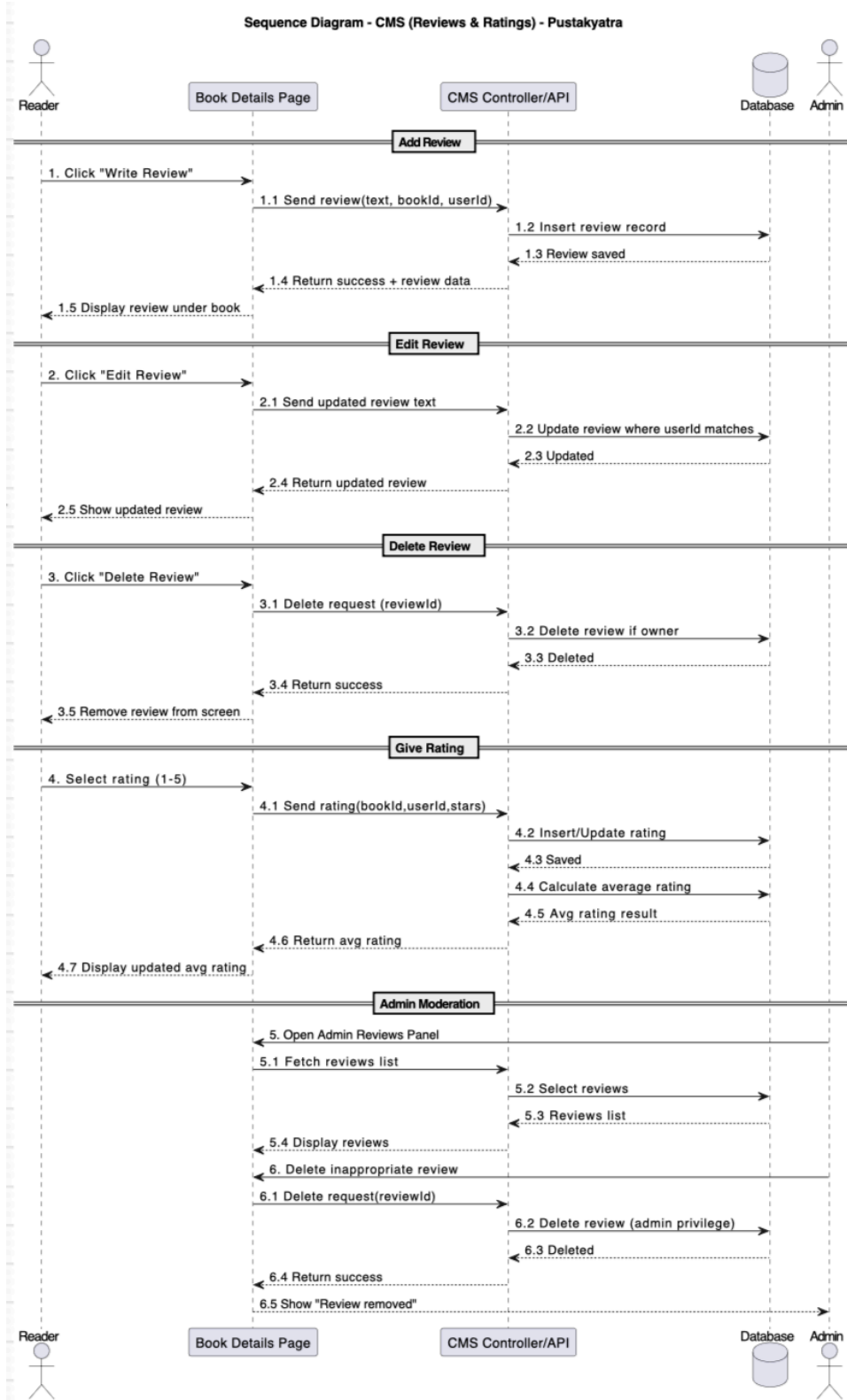


Figure 15: Sequence Diagram for CMS

The Comment Management System (CMS) in Pustakyatra is responsible for managing user interactions related to books, including reviews, ratings, and feedback. It allows readers to express opinions on books while enabling administrators to moderate and manage user-generated content to maintain quality and appropriateness.

Events Management System (EMS)

Req. Code	Requirement Type	Requirement Description	Use Case	MoSCoW
EMS-F-1.0	FR	Admin can create/add events with title, date, time, location/online link, description.	Add Event	Must Have
EMS-F-1.1	FR	Admin can update/edit events anytime.	Update Event	Must Have
EMS-F-1.2	FR	Admin can delete events (expired/wrong).	Delete Event	Must Have
EMS-F-2.0	FR	Admin can upload event banner/image for each event.	Upload Event Banner	Must Have
EMS-F-2.1	FR	Users can view event list and event details .	View Events	Must Have
EMS-F-2.2	FR	Users can like events .	Like Event	Could Have
EMS-F-2.3	FR	Users can share events (copy link/social share).	Share Event	Could Have
EMS-NF-1.0	NFR	System should load events quickly and show newest first.	View Events	Should Have
EMS-NF-1.1	NFR	Only Admin can add/edit/delete events (role-based access).	Admin Manage Events	Must Have
EMS-UR-1.0	UR	Events page should be easy: search/filter by date (upcoming/past).	Browse Events	Could Have

Table 6: SRS of EMS

Activity Diagram

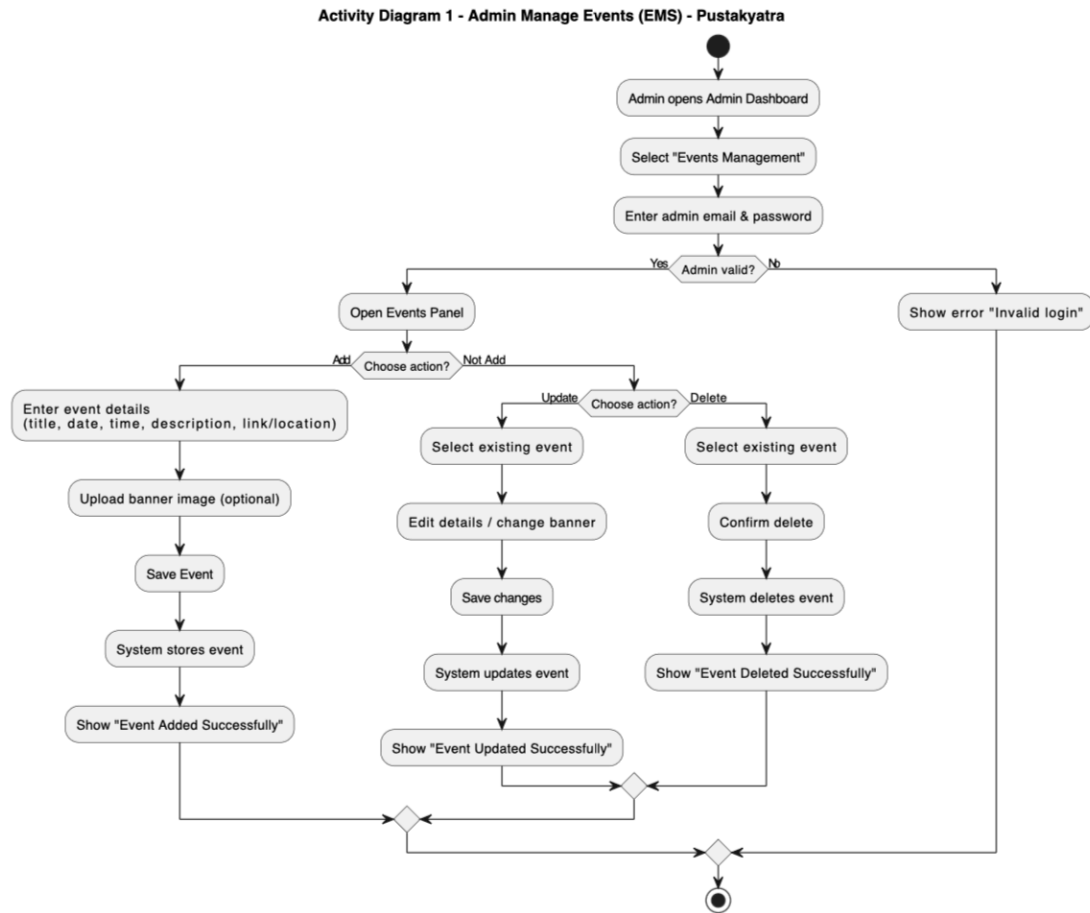


Figure 16: Activity Diagram (EMS)

Activity Diagram

Activity Diagram 2 - Reader View/Like/Share Events (EMS) - Pustakyatra

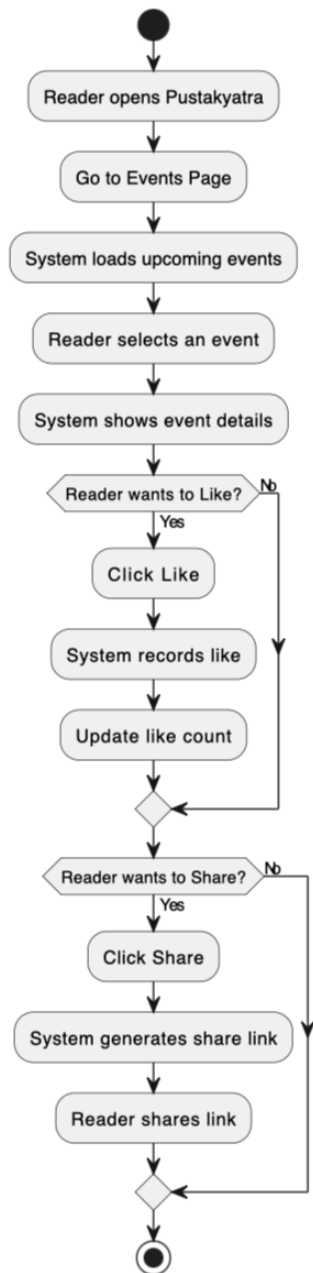


Figure 17: Activity Diagram 2(EMS)

USE CASE diagram

Use Case Diagram - EMS (Events) - Pustakyatra

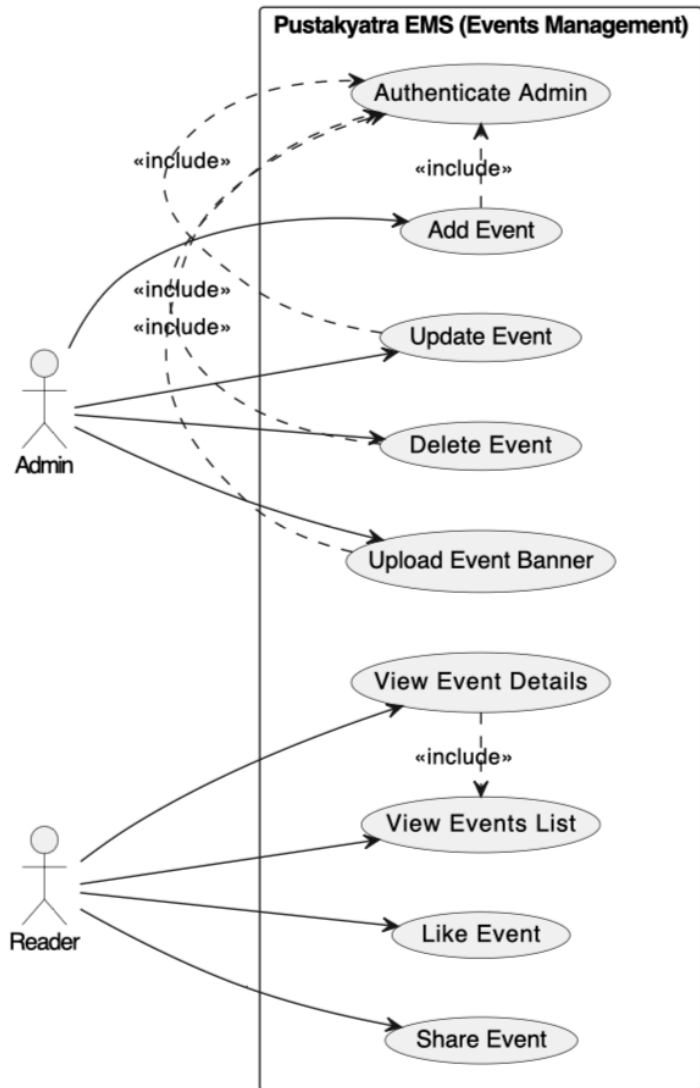


Figure 18: Use case of EMS

EMS – Sequence Diagram

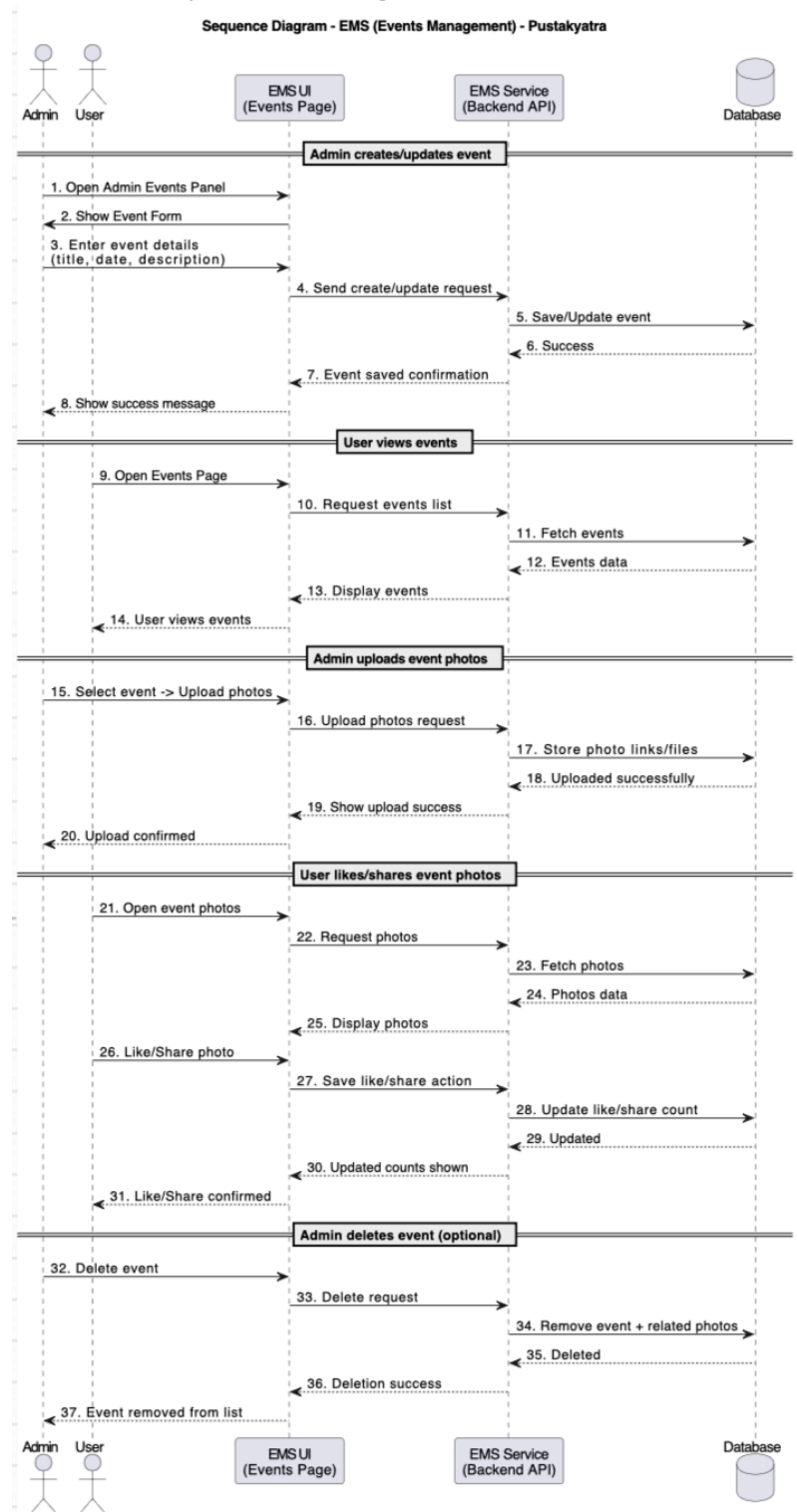


Figure 19: Sequence Diagram of EMS

NMS (News Management System)

Req. Code	Requirement Description	Use Case	MoSCoW Prioritization
NMS-F-1.0	Admin must be able to add news related to Pustakyatra such as new book arrivals, author announcements, offers, or system updates with required description and date.	Add news	Must Have
NMS-F-1.1	Admin must be able to update existing news information such as title, description, or image.	Update news	Must Have
NMS-F-1.2	Admin must be able to delete outdated or incorrect news from the system.	Delete news	Must Have
NMS-NF-1.3	The updated news must be visible to all readers using the Pustakyatra platform.	View news	Must Have
NMS-NF-1.4	System should display confirmation messages after adding, updating, or deleting news.	View status message	Should Have
NMS-NF-1.5	News should be displayed in chronological order with the latest news shown first.	View news list	Should Have

Table 7: SRS of News Management System

Activity Diagram

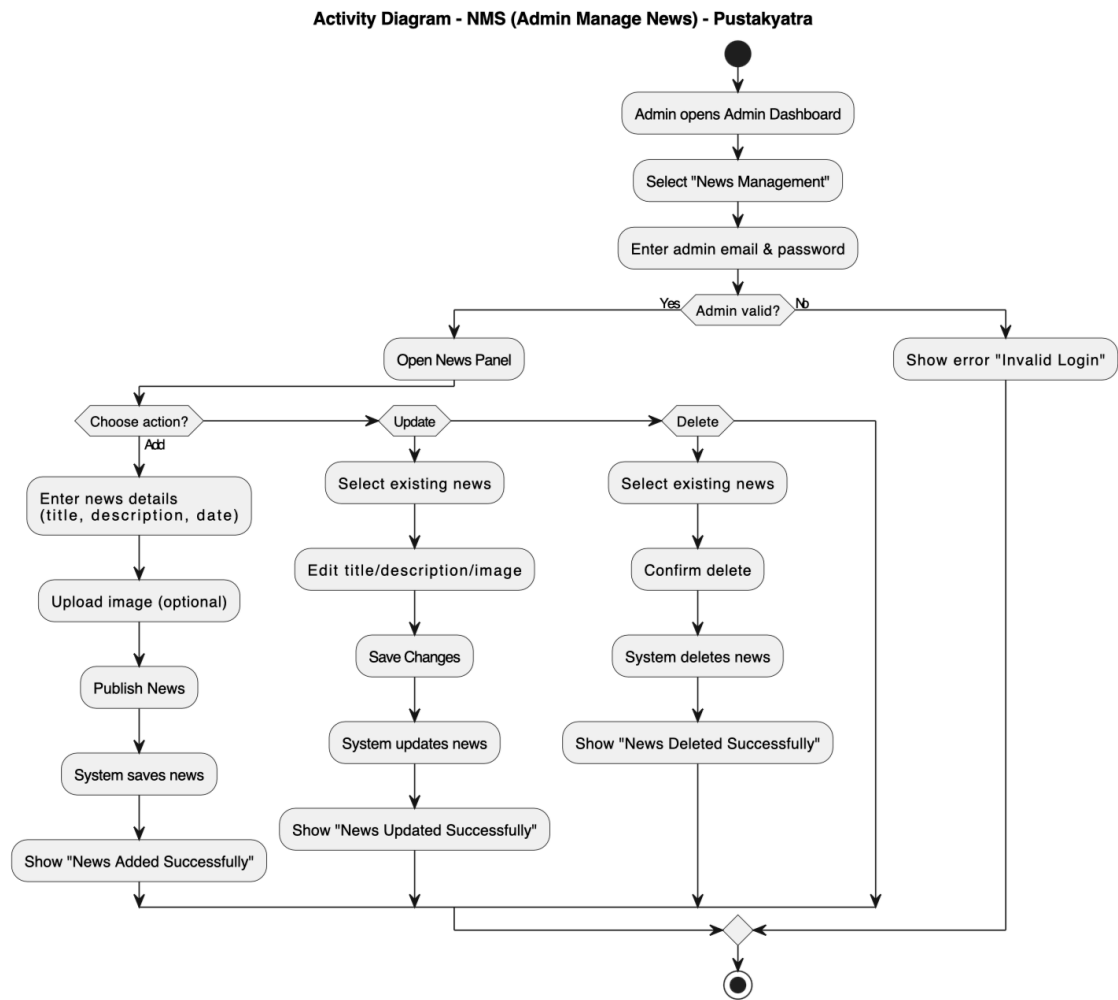


Figure 20: Activity Diagram of NMS

Use Case Diagram – NMS

Use Case Diagram - NMS (News Management System) - Pustakyatra

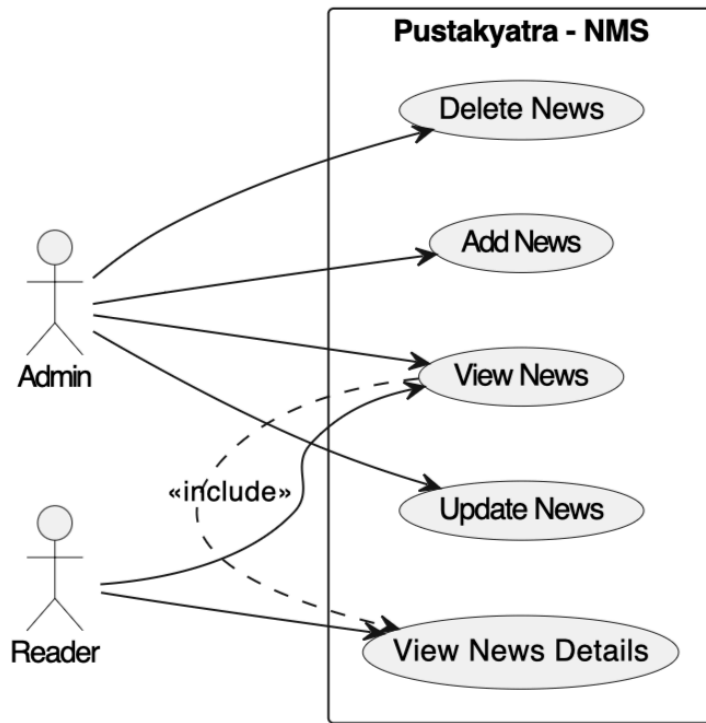


Figure 21: Use case of NMS

Sequence Diagram – NMS

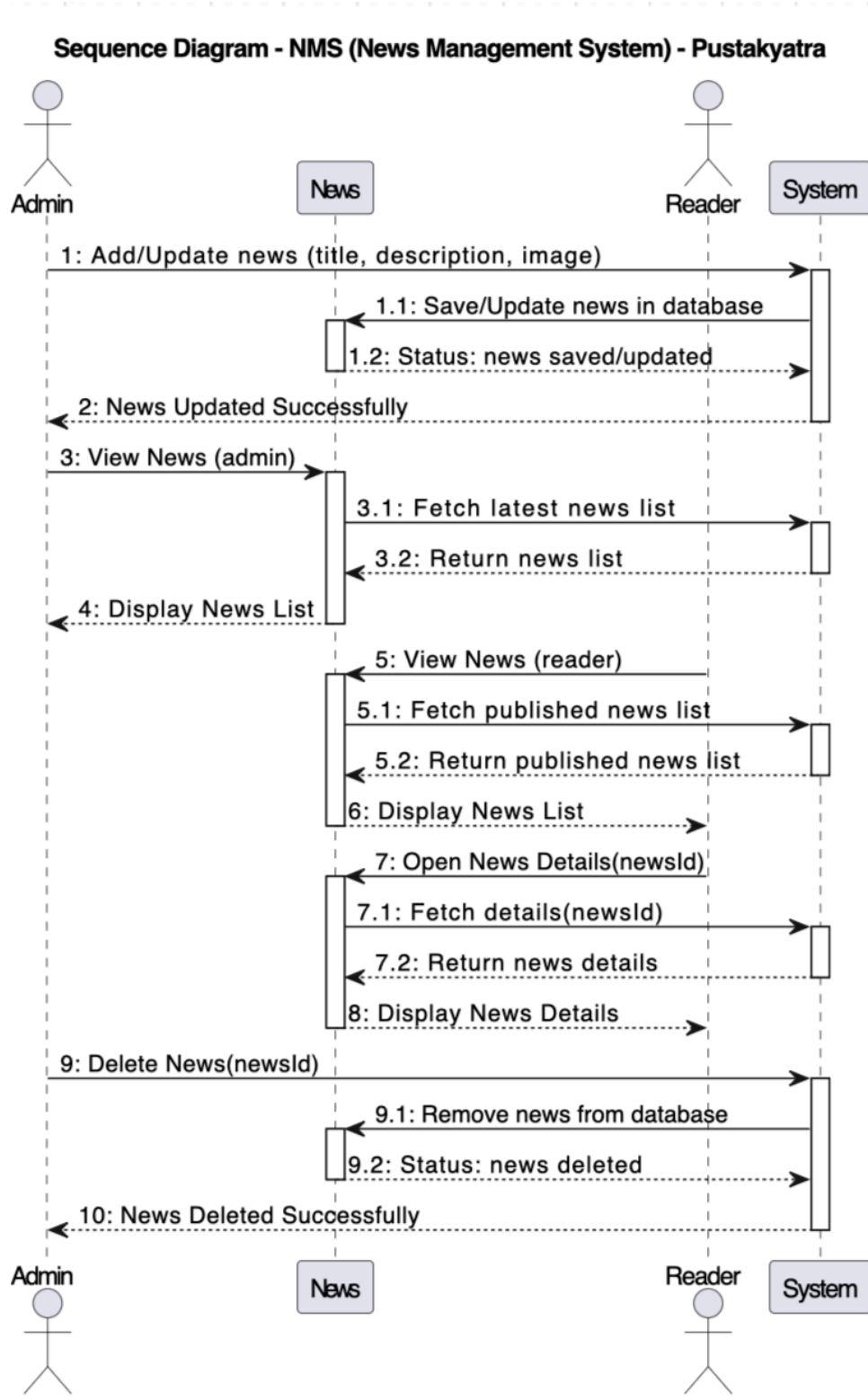


Figure 22: Sequence Diagram of NMS

Image Management System (IMS)

Req. Code	Requirement Description	Use Case	Moscow Prioritization
IMS-F-1.0	Admin must be able to upload images (book cover, banner, slider, author photo, news/event image) with required description/alt text .	Upload image	Must Have
IMS-F-1.1	Admin must be able to update/replace images (change file or description).	Update image	Must Have
IMS-F-1.2	Admin must be able to delete images that are outdated/wrong.	Delete image	Must Have
IMS-NF-1.3	Uploaded images must be visible to users in pages (Home, Catalog, About, News/Events).	View images	Should Have
IMS-NF-1.4	System should validate image type/size and store securely.	Validate upload	Could Have

Table 8: SRS of Image Management System

IMS Activity Diagram

Activity Diagram - IMS (Image Management System) - Pustakyatra

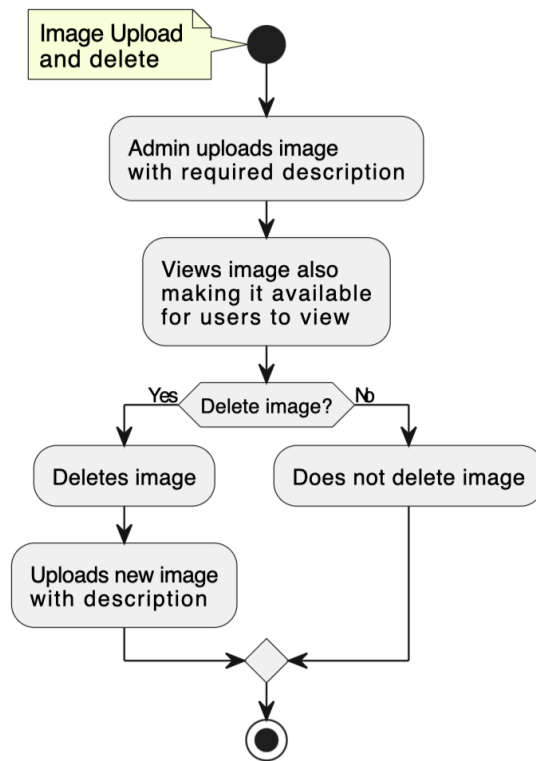


Figure 23: Activity Diagram OF IMS

Use Case Diagram(IMS)

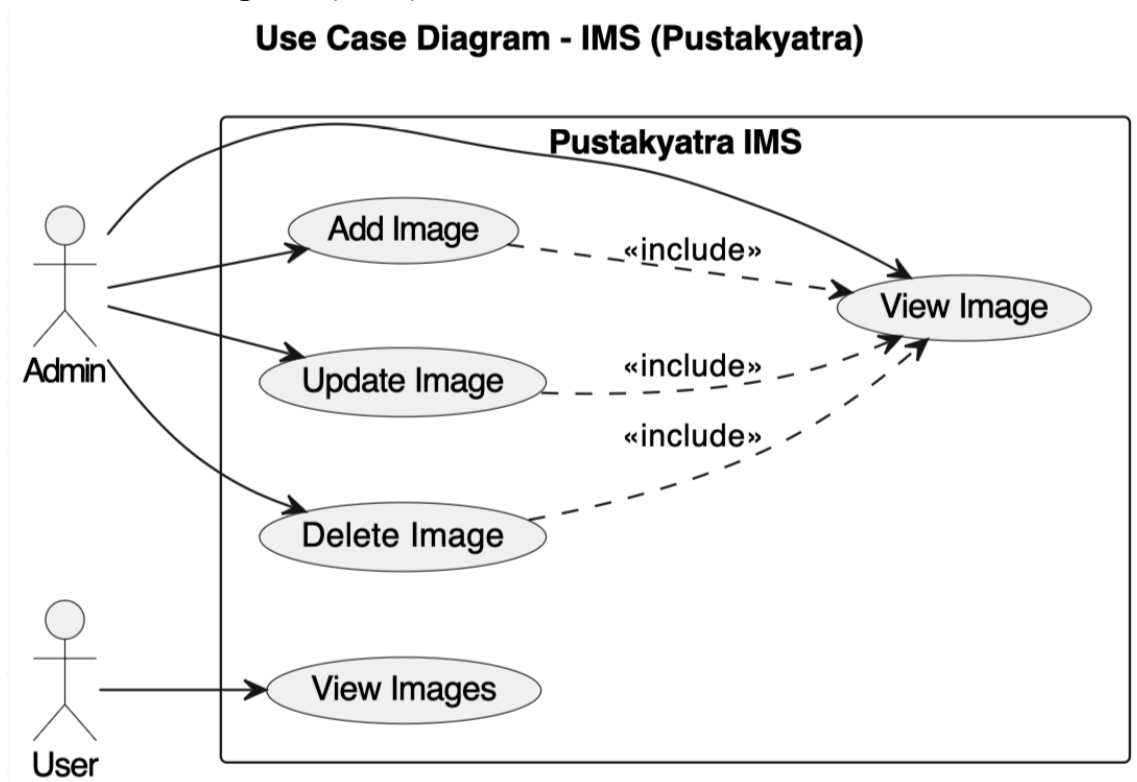


Figure 24: Use case Diagram of IMS

IMS Sequence Diagram

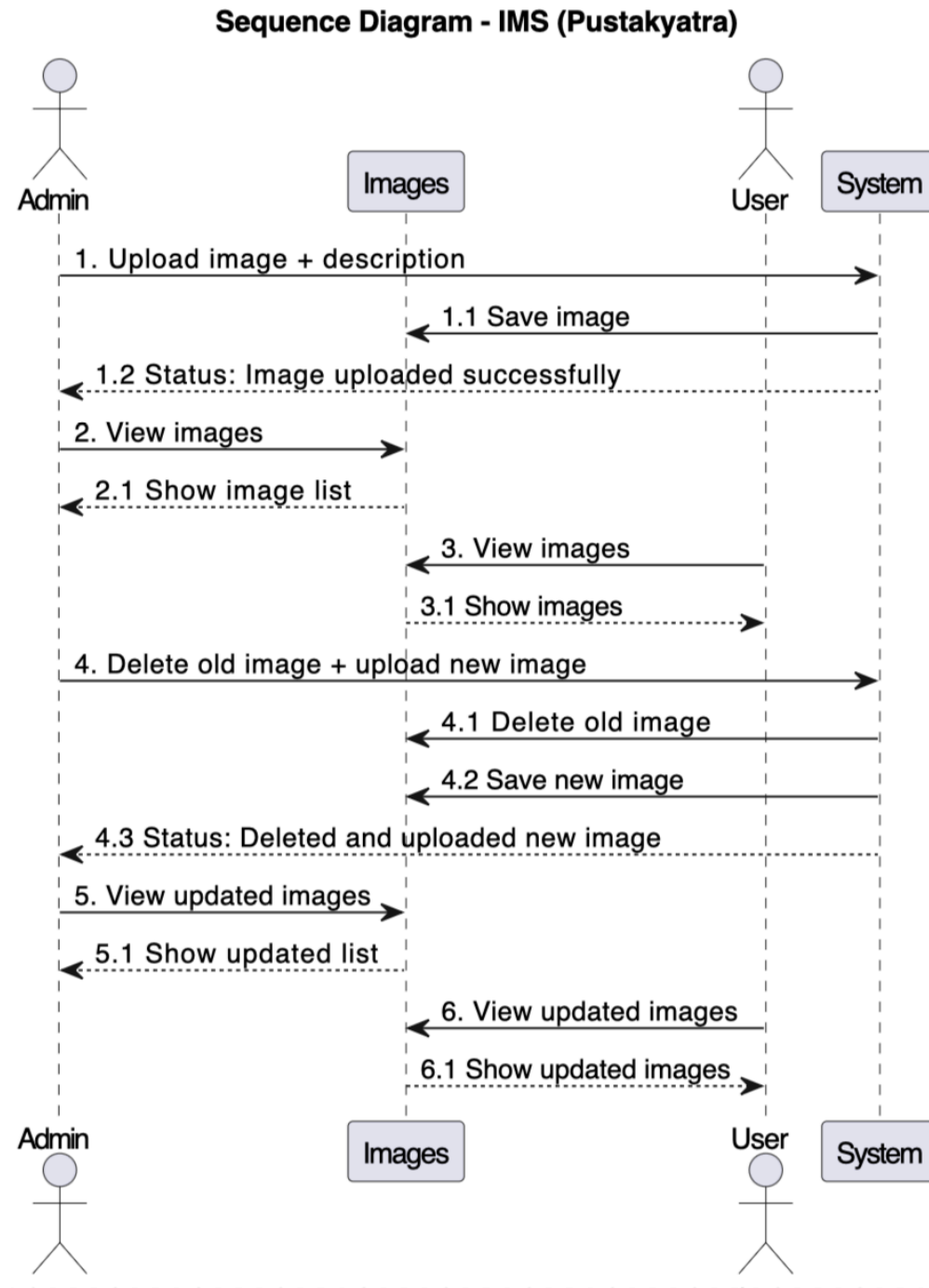


Figure 25: Sequence Diagram of IMS

Class Diagram - Pustakayatra (Clean Left → Right)

```

classDiagram
    class User {
        +int userId
        +string fullName
        +string email
        +Role role
    }
    class Book {
        +int bookId
        +string title
        +decimal price
        +decimal rentPrice
        +BookStatus status
    }
    class Payment {
        +int paymentId
        +decimal amount
        +PaymentStatus status
    }
    class Rental {
        +int rentalId
        +date startDate
        +date endDate
        +RentalStatus status
    }
    class AuthorProfile {
        +int authorId
        +string penName
    }
    class Admin {
        +int adminId
    }
    class ReaderProfile {
        +int readerId
        +string phone
    }
    class Bookmark {
        +int bookmarkId
        +string location
    }
    class ReadingSession {
        +int sessionId
        +float progressPercent
    }
    class Review {
        +int reviewId
        +int rating
        +string comment
    }
    class CoverImage {
        +int imageId
        +string imageUrl
    }
    class BookFile {
        +int fileId
        +string fileUrl
    }
    class Category {
        +int categoryId
        +string name
    }
    class Role {
        +READER
        +AUTHOR
        +ADMIN
    }
    class PaymentStatus {
        +PENDING
        +SUCCESS
        +FAILED
    }
    class BookStatus {
        +DRAFT
        +PUBLISHED
        +ARCHIVED
    }
    class RentalStatus {
        +ACTIVE
        +EXPIRED
        +CANCELLED
    }

    User "1" -- "0..*" Book
    User "1" -- "0..*" Payment
    User "1" -- "0..*" Rental
    User "1" -- "0..*" AuthorProfile
    User "1" -- "0..*" Admin
    User "1" -- "0..*" ReaderProfile
    User "1" -- "0..*" Bookmark
    User "1" -- "0..*" ReadingSession
    User "1" -- "0..*" Review
    Book "1" -- "0..*" Payment
    Book "1" -- "0..*" Rental
    Book "1" -- "0..*" BookFile
    Book "1" -- "0..*" Category
    Book "1" -- "0..*" Review
    Payment "1" -- "0..*" Book
    Rental "1" -- "0..*" Book
    Rental "1" -- "0..*" Payment
    Rental "1" -- "0..*" AuthorProfile
    Rental "1" -- "0..*" Admin
    Rental "1" -- "0..*" ReaderProfile
    Rental "1" -- "0..*" Bookmark
    Rental "1" -- "0..*" ReadingSession
    Rental "1" -- "0..*" Review
    BookFile "1" -- "0..*" Book
    Category "1" -- "0..*" Book
    Review "1" -- "0..*" Book
    Review "1" -- "0..*" User
    
```

The diagram illustrates the relationships between various entities in the Pustakayatra system. Key classes include User, Book, Payment, Rental, AuthorProfile, Admin, ReaderProfile, Bookmark, ReadingSession, Review, CoverImage, BookFile, and Category. Each class has specific attributes and roles. The diagram also includes enumerations for Role, PaymentStatus, BookStatus, and RentalStatus, which define the possible states and roles within the system.

Figure 26: Class Diagram of Pustakyatra

Entity-relationship Diagram

The ERD represents the database structure of Pustakyatra (E-Book Platform). It shows how users (reader, author, admin) interact with books through purchase or rental. The diagram also includes payment handling, reading progress, reviews, and notifications. Overall, it explains how all core features are connected within the system.

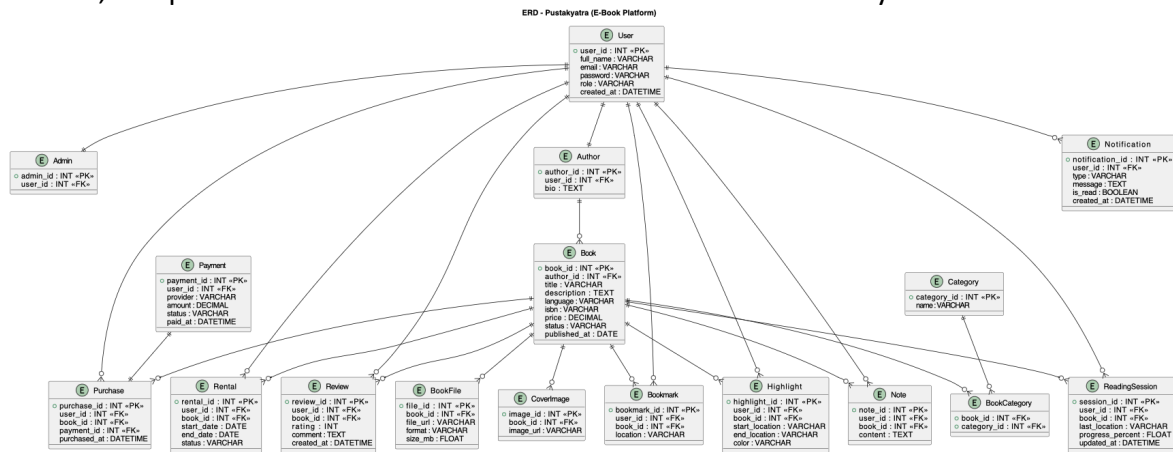


Figure 27: Entity-relationship Diagram of Pustakyatra

Wireframe:

Loginpage

Home Books About Us Contact Us Search [Login](#)

Login

Welcome back! Continue your reading journey with Pustakyatra.

Email*

Password*

☐ Remember me [Login](#)

Lost your password?
Don't have an account? [Register](#)

Figure 28: Wireframe for Login

Register

Welcome back! Continue your reading journey with Pustakyatra.

Full Name*

Email*

Password*

Confirm Password*

[Register](#)

Already have an account? [Login](#)

Figure 29: Wireframe for Register

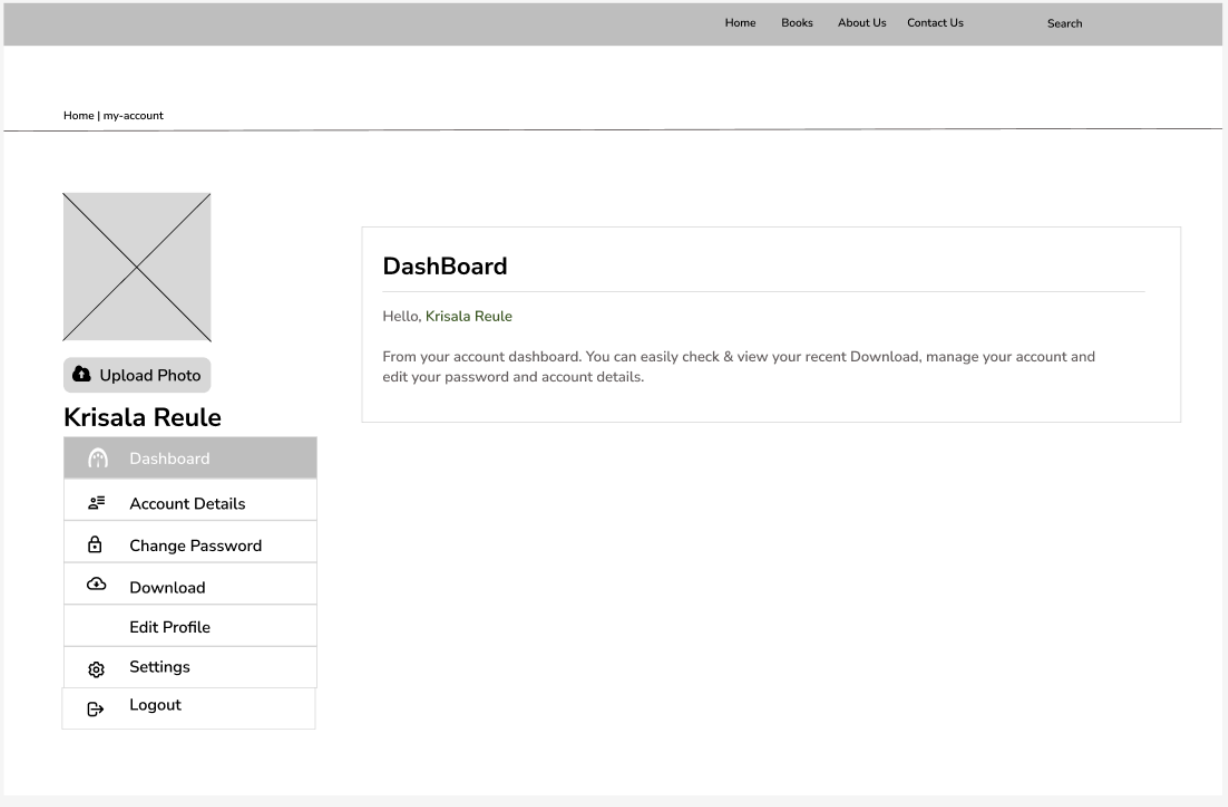


Figure 30: Wireframe for UserDashboard

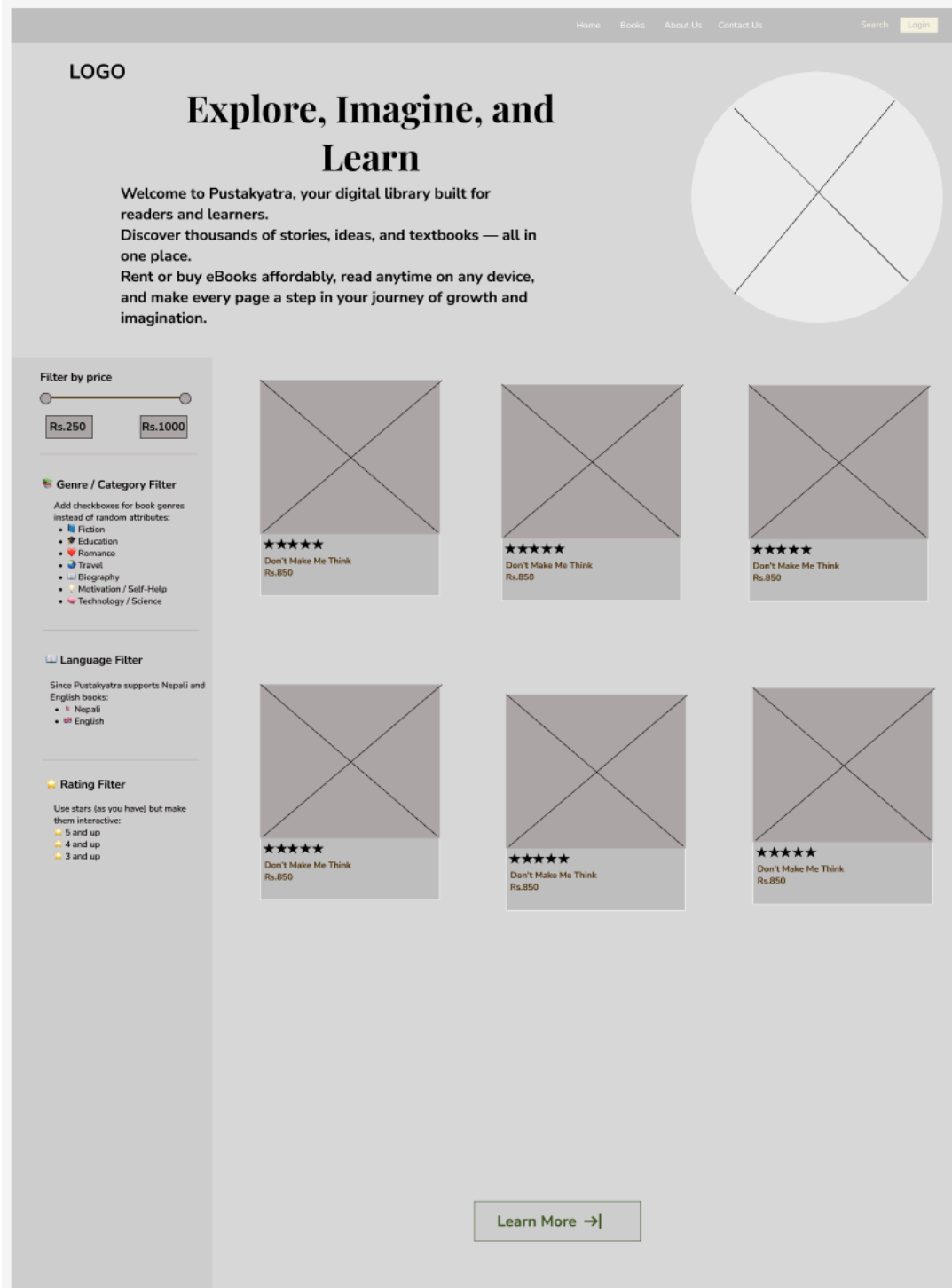


Figure 31: Wireframe for Homepage

UI-UX design of each page

Loginpage

Home Books About Us Contact Us Search [Login](#)

Login

Welcome back! Continue your reading journey with Pustakyatra.

Email*

Password*

☐ Remember me [Login](#)

Lost your password?
Don't have an account? [Register](#)

Figure 32: UI-UX for Login

Register

Welcome back! Continue your reading journey with Pustakyatra.

Full Name*

Email*

Password*

Confirm Password*

[Register](#)

Already have an account? [Login](#)

Figure 33: UI-UX for Register

[Home](#) | [my-account](#)[Upload Photo](#)**Krisala Reule**[Dashboard](#)[Account Details](#)[Change Password](#)[Download](#)[Edit Profile](#)[Settings](#)[Logout](#)

DashBoard

Hello, Krisala Reule

From your account dashboard. You can easily check & view your recent Download, manage your account and edit your password and account details.

Figure 34: UI-UX for UserDashboard

[Home](#) [Books](#) [About Us](#) [Contact Us](#) [Search](#) [Login](#)



Explore, Imagine, and Learn

Welcome to Pustakyatra, your digital library built for readers and learners.

Discover thousands of stories, ideas, and textbooks — all in one place.

Rent or buy eBooks affordably, read anytime on any device, and make every page a step in your journey of growth and imagination.



Filter by price

Genre / Category Filter

Add checkboxes for book genres instead of random attributes:

- ☐ Fiction
- ☐ Education
- ☐ Romance
- ☐ Travel
- ☐ Biography
- ☐ Motivation / Self-Help
- ☐ Technology / Science

Language Filter

Since Pustakyatra supports Nepali and English books:

- ☐ Nepali
- ☐ English

Rating Filter

Use stars (as you have) but make them interactive:

- ☐ 5 and up
- ☐ 4 and up
- ☐ 3 and up



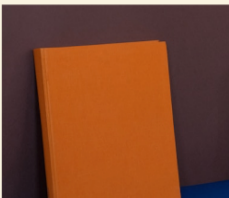
★★★★★
Don't Make Me Think
Rs.850



★★★★★
Don't Make Me Think
Rs.850



★★★★★
Don't Make Me Think
Rs.850



★★★★★
Don't Make Me Think
Rs.850



★★★★★
Don't Make Me Think
Rs.850



★★★★★
Don't Make Me Think
Rs.850

Figure 35: UI-UX for Homepage